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6

The OS and the Filesystem

C06: System & AI Essentials

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OS Next Role: Manage Files & Storage

How does the OS organise everything stored on your computer?

Everything is, well, in the room somewhere...





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If you were looking for something, in which room would be easier to find the thing?

What if everything in a computer was randomly stored?



How would the system find the right file ?

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When everything has a place, it becomes easier to find and manage.

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The OS has an organised system...

```
› ls /
Applications  Library     Users
bin           opt         usr
cores         private     var
dev           sbin        Volumes
etc           System
home          tmp
```

This organised structure is called the filesystem.

Everything on your computer is stored somewhere under the root location (/).

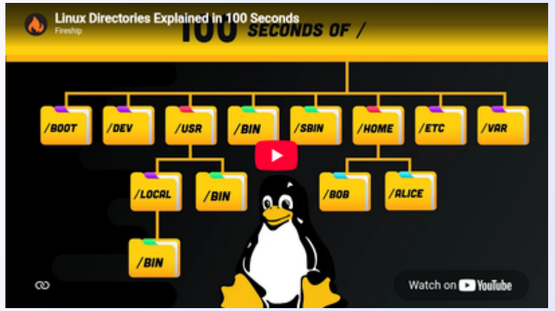
Every place has a purpose



Directory	Purpose
/bin	Programs
/etc	Configuration / Settings
/dev	Devices represented by files
/tmp	Temporary files
/Users	User data
/Applications	Graphical apps on macOS

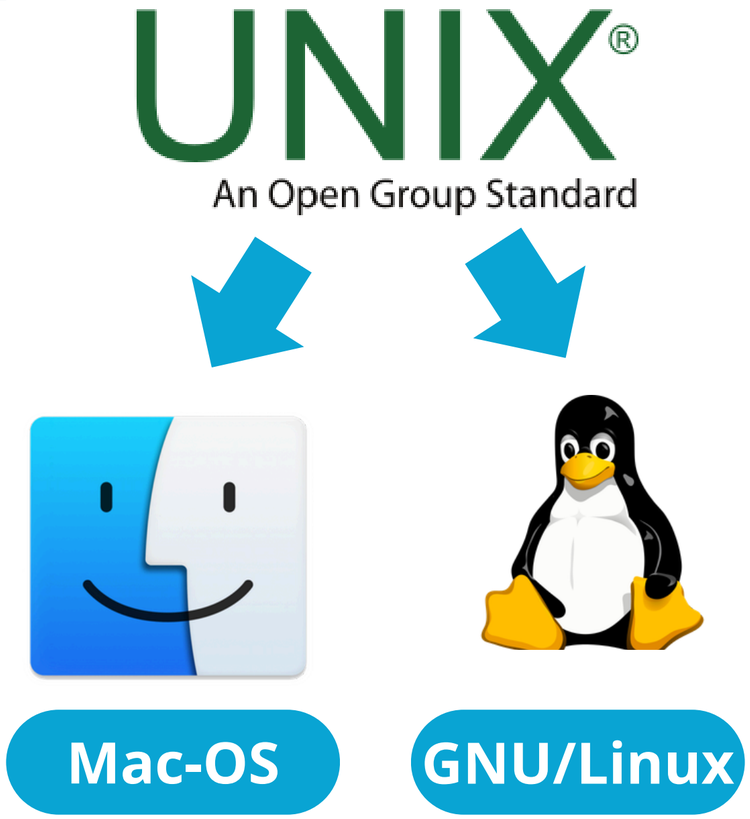
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This is inspired from the UNIX OS
Let’s watch a video. He is helping to name directories faster than I can.



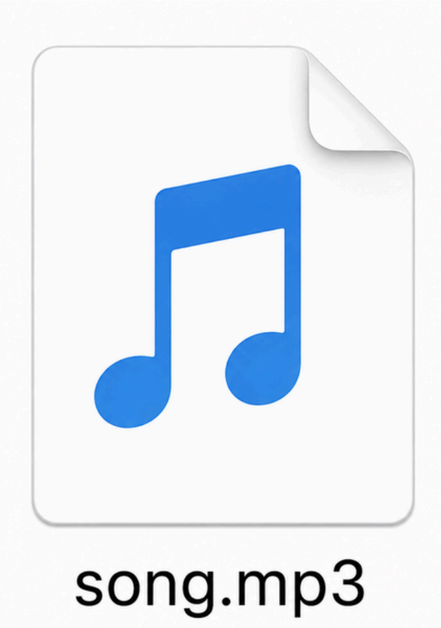
This is inspired from Unix

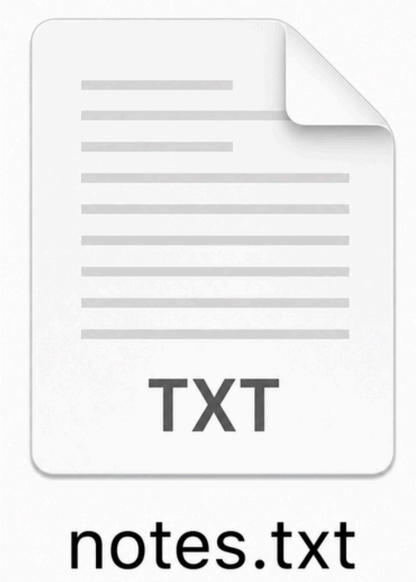
- **Unix** was built around one powerful idea.
- ***Everything is a file:*** one common way to interact with many **system resources**.



Its ideas later influenced operating systems such as Linux and macOS.

What does Everything is a file mean?

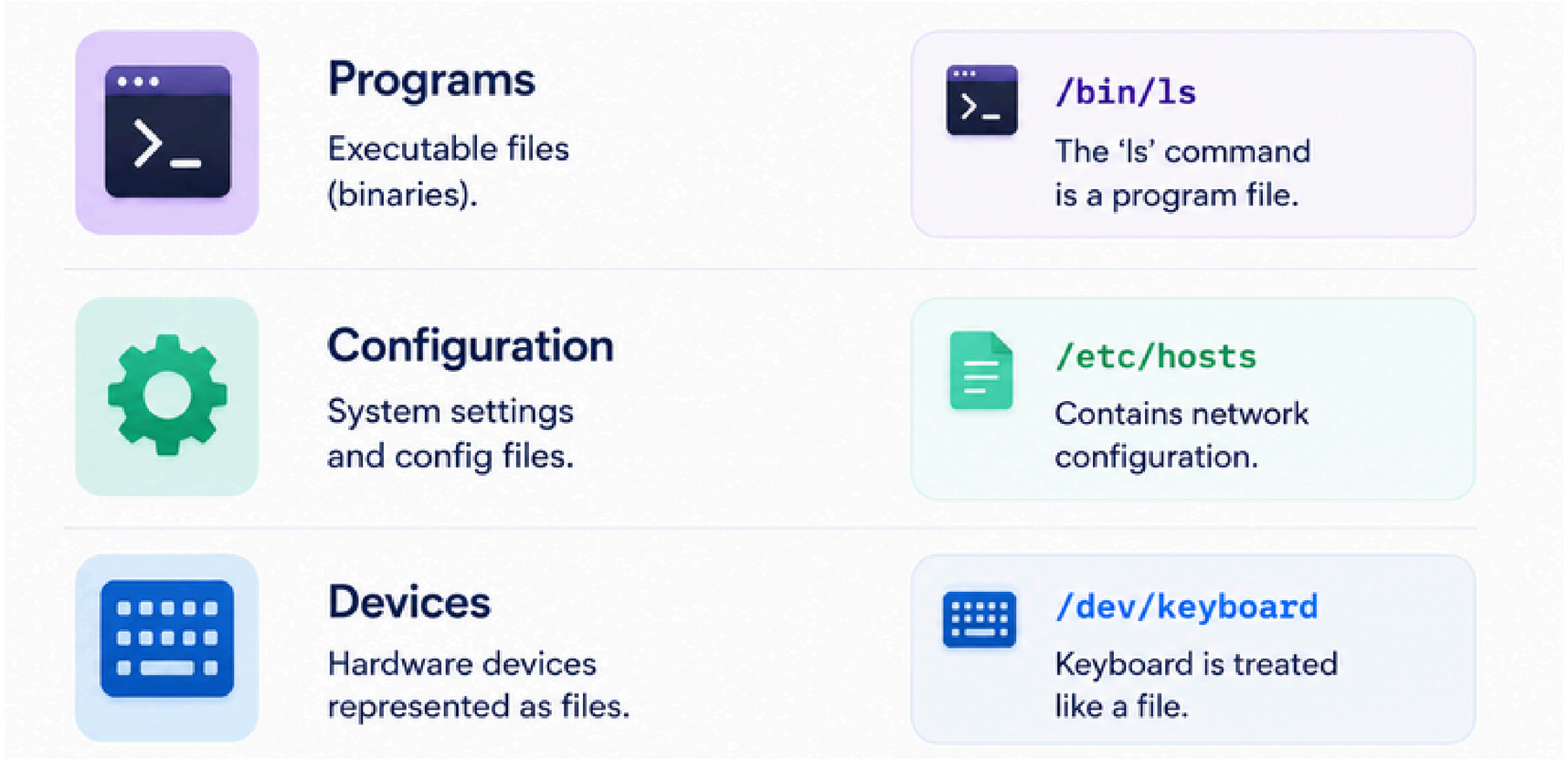






Normal Files

Unix also treats these as files



Commands we ran, are files

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```
› ls /bin
[           csh         echo        ksh         mkdir       realpath    stty        wait4path
bash        dash        ed          launchctl   mv          rm          sync        zsh
cat         date        expr        link        pax         rmdir       tcsh
chmod       dd          hostname    ln          ps          sh          test
cp          df          kill        ls          pwd         sleep       unlink
```

Program on disk = file
Program in execution = process

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Demo on Ubuntu UTM for /dev/pts & /dev/input

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What actually makes a file executable?

Activity: Let’s create a simple program

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1) Create a new file called hello.txt
2) Add a simple instruction inside the file
3) Try to execute the file

Trying to run the file

```
› mkdir terminal-practice
› cd terminal-practice
› touch hello.txt
› echo "echo Hello from a script" > hello.txt
./hello.txt
zsh: permission denied: ./hello.txt
 ~/terminal-practice                                                    126 ✘  02:45:50 PM
```

Command 1: ***mkdir terminal-practice***
Command 2: ***cd terminal-practice***
Command 3: ***touch hello.txt***
Command 4: ***echo "echo Hello from a script" > hello.txt***
Command 5: ***./hello.txt***

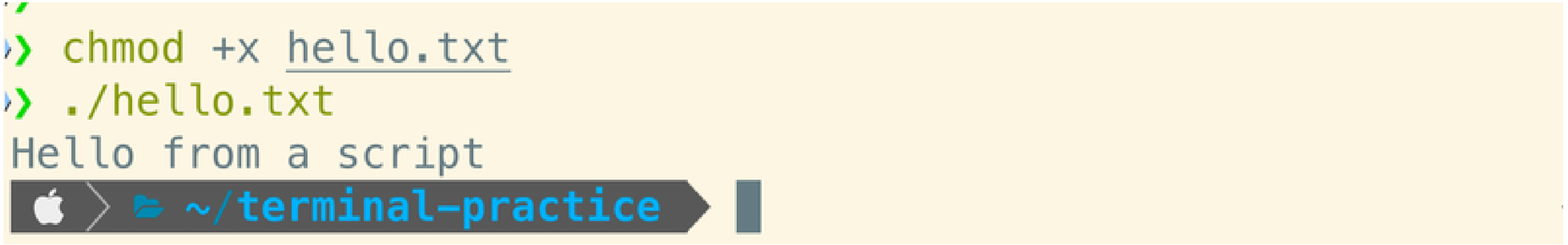
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For a file to run, the OS checks two things.

1. Does the user have permission to run/execute it?
2. Does it contain executable instructions?

chmod

The chmod command in Linux is used to change file or directory permissions..



Command 1: ***mkdir terminal-practice***
Command 2: ***cd terminal-practice***
Command 3: ***touch hello.txt***
Command 4: ***echo "echo Hello from a script" > hello.txt***
Command 5: ***./hello.txt***
Command 6: ***chmod +x hello.txt***
Command 7: ***./hello.txt***

What if change the filename? will it work?

The filename doesn’t even matter!

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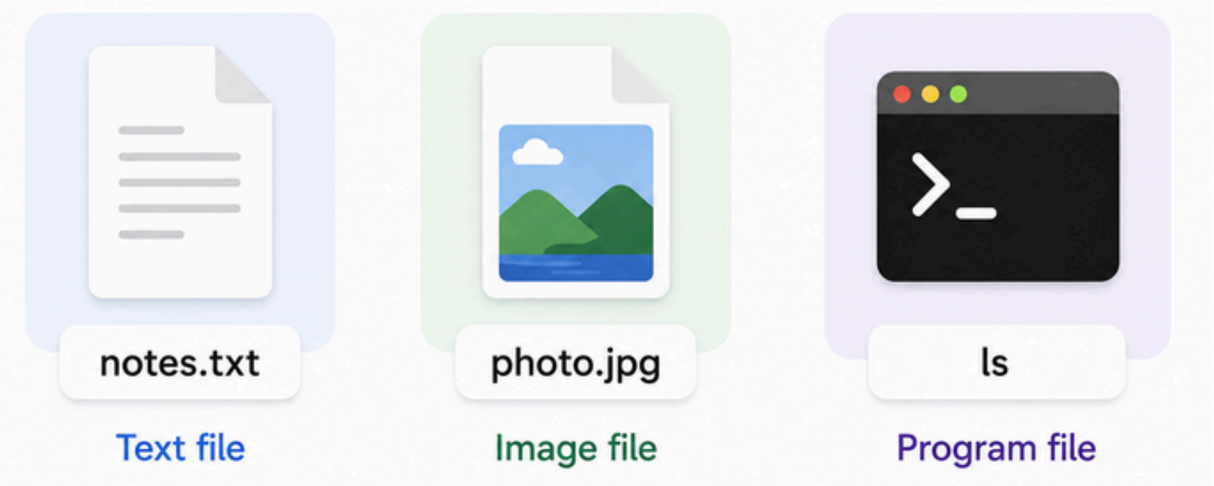
```
❯ mv hello.txt run_me.banana
❯ ./run_me.banana
Hello from a script
 ~/terminal-practice
```

Command 1: ***mkdir terminal-practice***
Command 2: ***cd terminal-practice***
Command 3: ***touch hello.txt***
Command 4: ***echo "echo Hello from a script" > hello.txt***
Command 5: ***./hello.txt***
Command 6: ***chmod +x hello.txt***
Command 7: ***./hello.txt***
Command 8: ***mv hello.txt run_me.banana***
Command 9: ***./run_me.banana***

Not every file contains instructions

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- notes.txt → stores text
- photo.jpg → stores image data
- /bin/ls → stores machine-runnable instructions



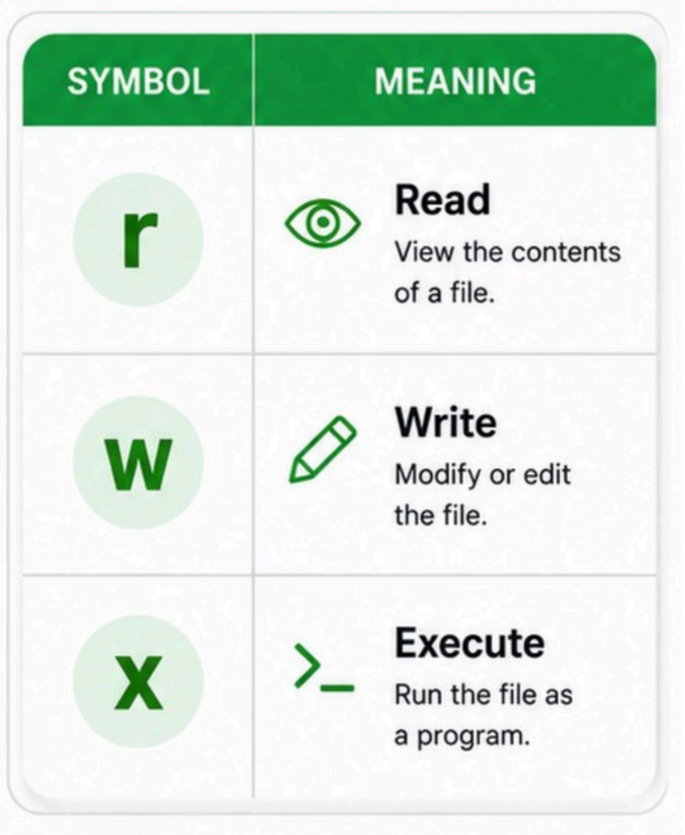
A file can store many kinds of data, but only some files contain instructions that can be executed.

Understanding File Permissions

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Each file has **three** types of **permissions:**

- **Read (r)** – Can view or open the contents of the file
- **Write (w)** – Can modify or edit the file
- **Execute (x)** – Can run the file as a program



Permissions are assigned for Owner, Group, and Others.

Default Permissions

Default Read/Write Permissions:

- When a new file is created with touch hello.txt, the owner already has read and write permissions (rw-).
- This is why hello.txt can be viewed or edited immediately.

Execute Permission is Separate:

- By default, a file is not executable (-rw-r--r--) to prevent accidental execution.
- Even if the file contains instructions, the OS will not run it unless the execute bit is set.
- Ensures system security and prevents accidental or unsafe execution.



File Permissions & Owners



Students





Teachers

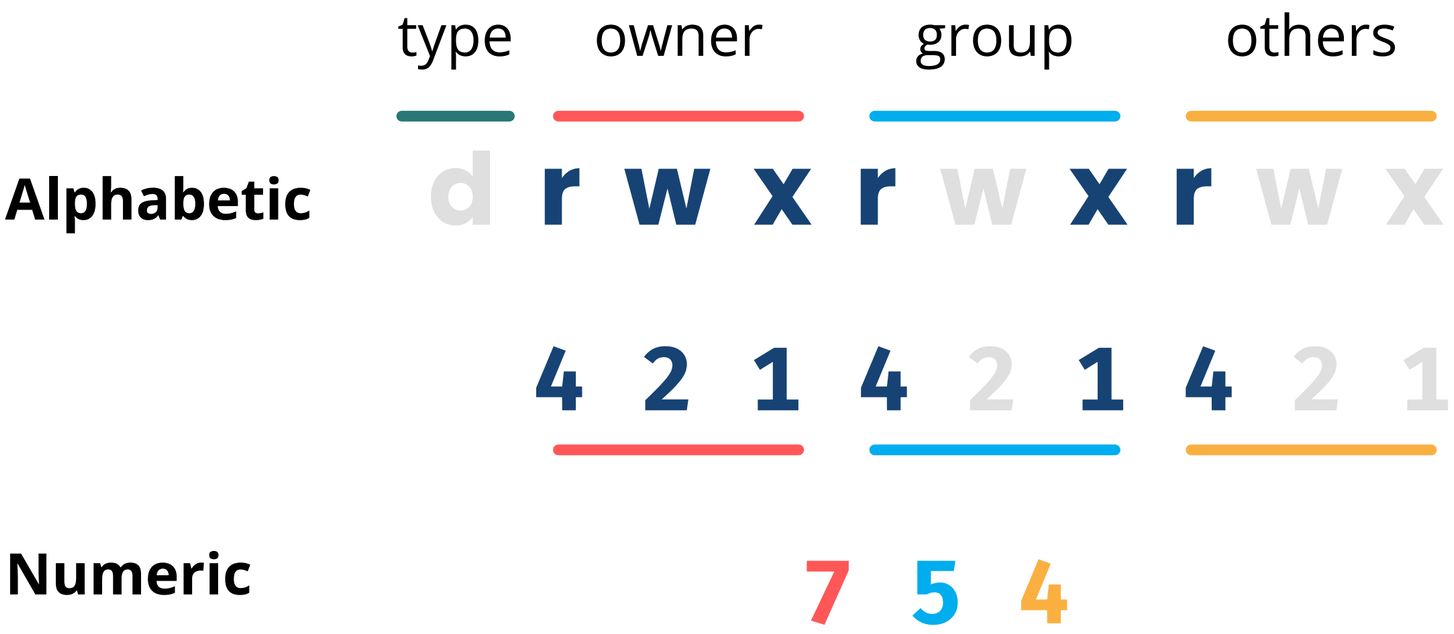
```
Owner: Karan Sir
Owner's Group: Teachers
```

Owner can: R W
Group can: R
Others: R

Alphabetic form: RW-R--R--
Numeric form: 644

Permission Flags

Permissions are represented by a set of characters (e.g., rwxr-xr--) or numeric values (e.g., 755), defining who can read, modify, or execute the file.



Taking a closer look at ls -l

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```
❯ ls -l
total 8392
drwx------@   6 tanmay  staff      192  2 Jul 12:54 Applications
-rw-r--r--@   1 tanmay  staff     2443  8 Jan  2026 aws_test.py
drwxr-xr-x@   3 tanmay  staff       96  2 Sep  2025 CascadeProjects
drwxr-xr-x    6 tanmay  staff      192  6 Sep 12:37 Classroom_Discussions
drwxr-xr-x@  12 tanmay  staff      384 17 Jul 10:49 code
drwx------@  80 tanmay  staff     2560  6 Sep 15:31 Desktop
drwx------@  34 tanmay  staff     1088  6 Sep 15:31 Documents
drwx------@ 306 tanmay  staff     9792  6 Sep 11:57 Downloads
-rw-r--r--    1 tanmay  staff       48 21 May 11:21 file.txt
```

When we run **ls** with the **-l** option, we see the long-form output shown alongside.

We can observe a column with values such as '-rw-r--r--' for certain files. These values represent the **permissions**.

Permission Examples

```
r = 4      rwx = 7        chmod 755  →  scripts, programs
w = 2      r-x = 5        chmod 644  →  normal files
x = 1      r-- = 4        chmod 600  →  private
           --- = 0        chmod 640  →  /etc/shadow
```

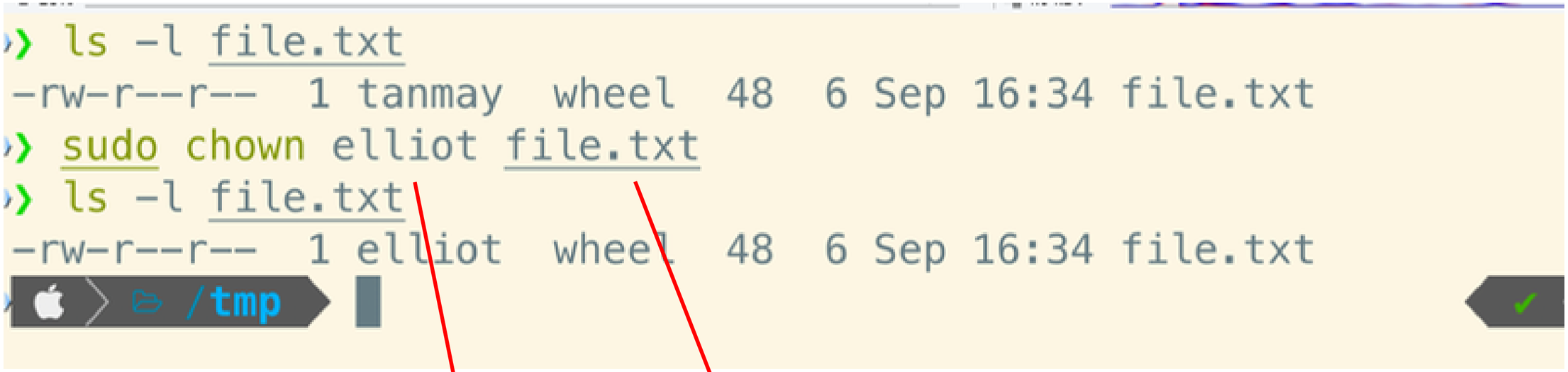
- **echo "private data" > secret.txt**
- **chmod 600 secret.txt**
- **ls -la secret.txt**

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But what if another user needs to own or manage this file? How do we change the owner safely?

chown Transfer File Ownership

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New owner filename

chown changes the owner of a file. Useful sometimes...

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The command ls from any folder and it worked.

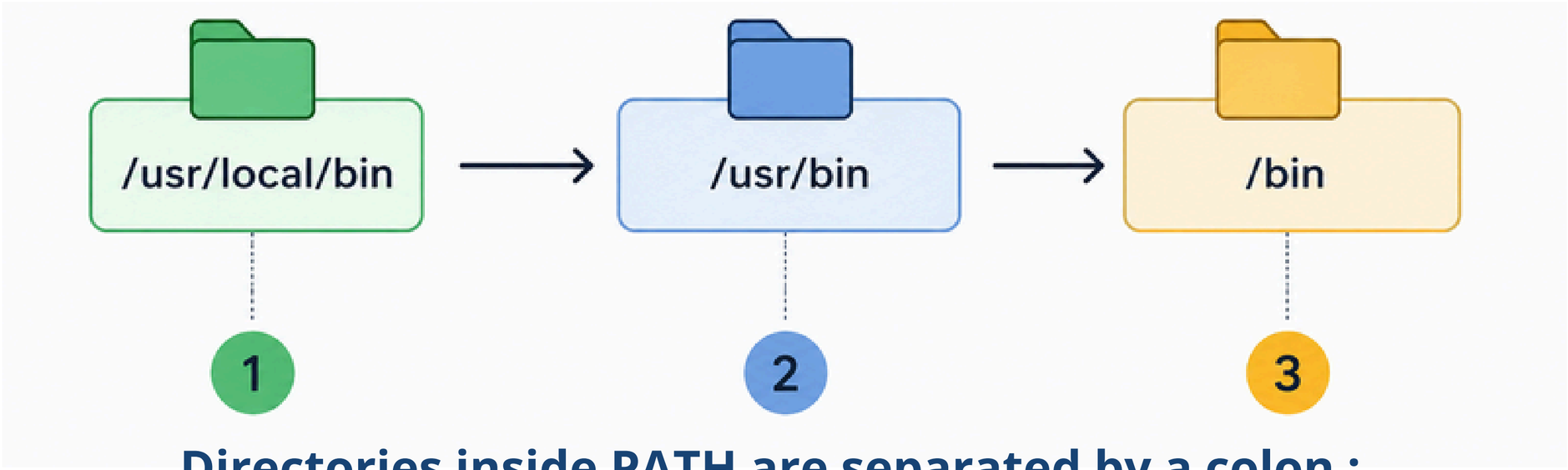


Why? How does the terminal know where ls lives?

PATH: The Shell’s Search List

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- PATH is an environment variable containing a list of directories.
- The shell searches these directories from left to right until it finds the command.



Directories inside PATH are separated by a colon :

PATH="/usr/local/bin:/usr/bin:/bin"

Reading Your PATH

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```
› echo $PATH
/usr/bin:/bin:/usr/sbin:/sbin
```



If the command exists in one of these directories, it runs; otherwise, you get command not found.

What Happens If PATH is Empty?

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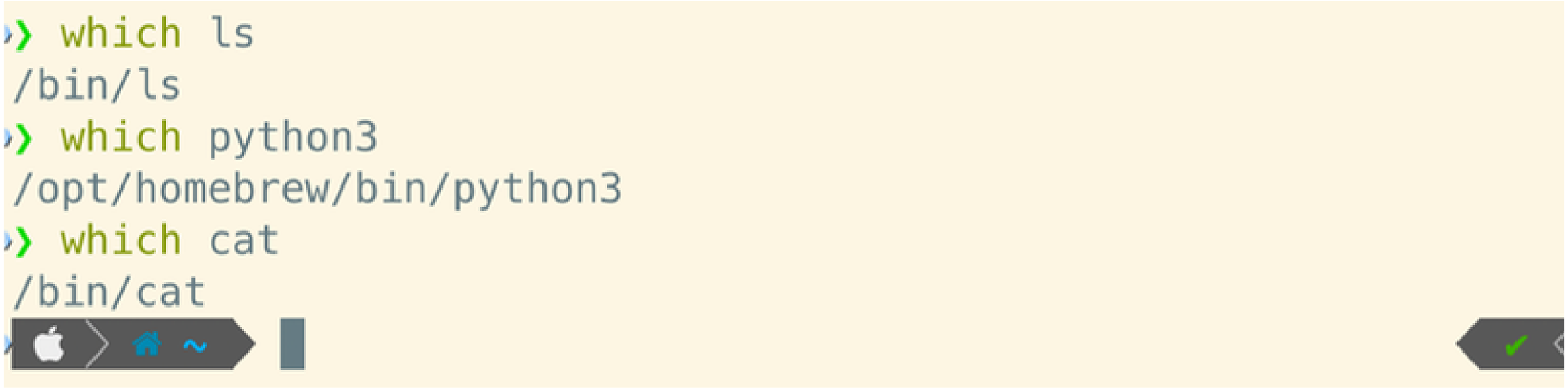
```
› PATH=""
› ls
zsh: command not found: ls
› cat
zsh: command not found: cat
› /bin/ls
Applications                    mypod.yaml
aws_test.py                     newdeployment.yaml
CascadeProjects                 node_modules
Classroom_Discussions           output.json
```

ls fails because the shell cannot search for it.

/bin/ls works because its exact location is provided.

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Now let us trace how the shell finds ls



You can find ls using which that relies on searching the directories in PATH

Summary

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Filesystem stores programs as files

↓

Permissions decide who may execute them

↓

PATH helps the shell locate commands

↓

which command will tell you where the location of executable is

Feedback

Now it's time to give your valuable feedback



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Thank You!